

MVCL-DTI: Predicting Drug–Target Interactions Using a Multiview Contrastive Learning Model on a Heterogeneous Graph

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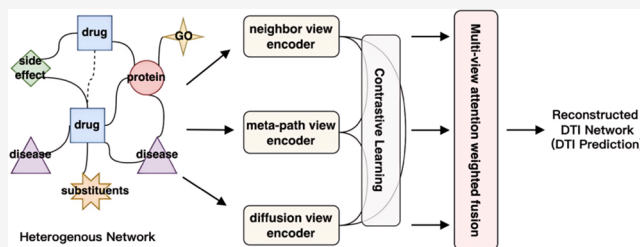


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ABSTRACT: Accurate prediction of drug–target interactions (DTIs) is pivotal for accelerating the processes of drug discovery and drug repurposing. MVCL-DTI, a novel model leveraging heterogeneous graphs for predicting DTIs, tackles the challenge of synthesizing information from varied biological subnetworks. It integrates neighbor view, meta-path view, and diffusion view to capture semantic features and employs an attention-based contrastive learning approach, along with a multiview attention-weighted fusion module, to effectively integrate and adaptively weight the information from the different views. Tested under various conditions on benchmark data sets, including varying positive-to-negative sample ratios, conducting hard negative sampling experiments, and masking known DTIs with different ratios, as well as redundant DTIs with various similarity metrics, MVCL-DTI exhibits strong robust generalization. The model is then employed to predict novel DTIs, with a particular focus on COVID-19-related drugs, highlighting its practical applicability. Ultimately, through features visualization and computational properties analysis, we've pinpointed critical elements, including Gene Ontology and substituent nodes, along with a proper initialization strategy, underscoring their vital role in DTI prediction tasks.



1. INTRODUCTION

In most drug design projects, the first step in the drug discovery pipeline is to identify ligand molecules that bind with high affinity to a specific protein target. These molecules can be further developed into drug-like compounds, as drugs exert their therapeutic effects by interacting with proteins.¹ Experimental methods for identifying drug molecules, such as high-throughput screening, can be both time-consuming and expensive. In contrast, computational methods for DTI prediction can significantly reduce the required resources, time, and cost, as well as minimize the need for biological wet-lab studies to screen new therapeutic candidates.

Computational DTI prediction methods can be broadly classified into docking-based methods, ligand-based methods, and machine learning-based methods.² Docking-based methods are computationally intensive and require precise measurements of protein 3D structures.^{3,4} Ligand-based methods often follow the similarity principle,⁵ which states that structurally similar compounds tend to have similar biological activities. While these methods do not require detailed structural information on the protein, their accuracy heavily depends on the availability and quality of known DTI data. Machine learning-based methods are increasingly popular for DTI prediction due to their ability to handle large-scale and complex biological data. These methods utilize various algorithms to learn patterns and relationships

from known interactions and use this knowledge to predict new interactions.

Existing DTI prediction models largely rely on machine learning methods, and graph neural networks (GNNs) are widely used. By representing drug molecules and proteins as graphs, with nodes representing atoms or amino acids and edges representing chemical bonds, GNNs can capture the functional structures and chemical properties of drugs and proteins. The learned representations facilitate the prediction of DTIs. NetLapRLS⁶ constructs drug and target similarity matrices based on chemical structure, genomic data, and other features. EADTN⁷ introduces an innovative feature adaptation technique to automatically extract local weights of drugs and targets. It also employs clustering-enhanced parameter fine-tuning to address the issue of false negatives, enhancing its reliability in drug discovery. GCN-DTI⁸ builds drug–protein pair (DPP) networks and transforms the edge prediction problem into a DPP classification problem. However, constructing homogeneous graphs offers a detailed microlevel analysis of interactions, yet it

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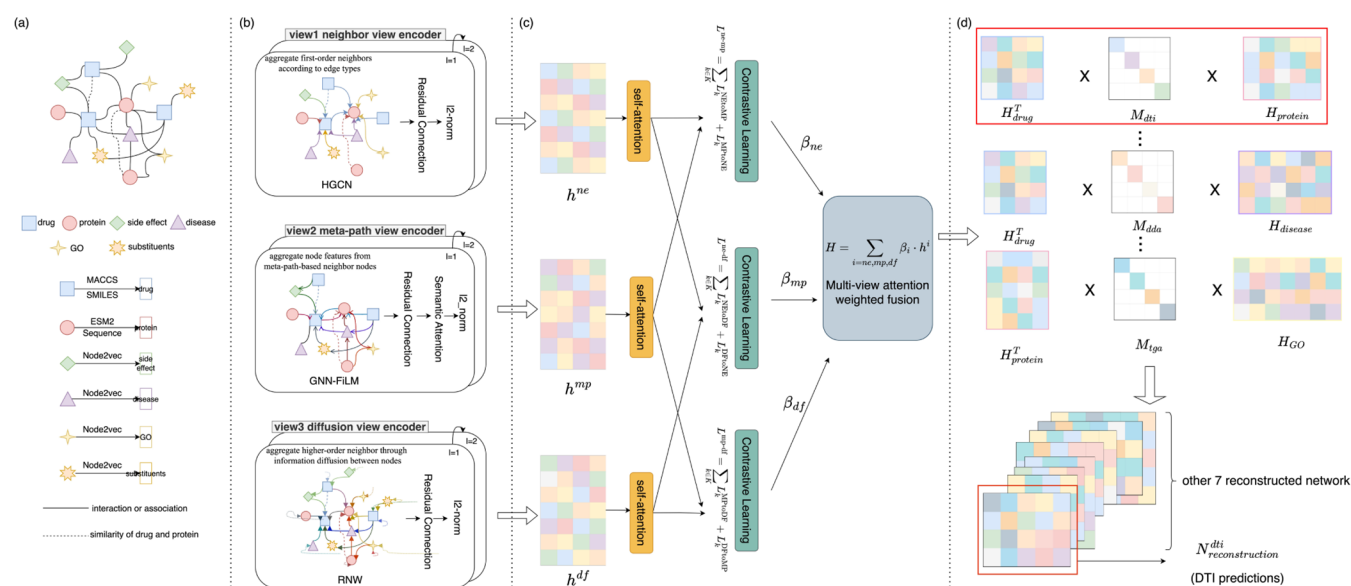


Figure 1. Architecture of MVCL-DTI. MVCL-DTI mainly consists of four parts: (a) Construction of the heterogeneous graph. This graph includes six types of nodes: drugs, proteins, diseases, side effects, substituents, and GO terms. The interactions, associations, or similarities between them are represented as different types of edges. (b) Representation learning from three distinct views. The neighbor view encoder captures 2-hop neighbor information, the meta-path view encoder derives features from meta-path-based neighbors, and the diffusion view encoder captures higher-order neighbor information via random walks. (c) Integration of multiview information through contrastive learning. An attention weighted fusion mechanism is used to effectively integrate representations from the three views. (d) Reconstruction of the heterogeneous network. The node representations from the neighbor view are used to reconstruct the original network, with the resulting DTI network serving as the final prediction outcome.

restricts the integration of complex biological contexts, such as disease correlations and drug side effects, which are essential for a comprehensive understanding.

Further, introducing heterogeneous graphs, which include multiple types of nodes and edges, enriches the semantic context compared to homogeneous graphs. Heterogeneous graphs take proteins, drugs, diseases, etc., as nodes, and interactions and associations as edges, allowing the model to learn comprehensive topological information and complex interactions in the biological network. For heterogeneous graphs specifically involving drugs and proteins, the DPI model⁹ presents a novel framework by integrating global and local features of nodes in the drug–protein bipartite graph for efficient DPI inference. IGRLDTI¹⁰ employs a heterogeneous biological interaction network (HBIN) by integrating the biological knowledge of drugs and targets with their interactions and a node-dependent local smoothing strategy to adaptively determine the propagation depth for each biomolecule, mitigating oversmoothing issues and improving drug and target representations. In the meantime, most GNNs target homogeneous graphs, where each layer aggregates information from single-hop neighbors. Applying them to heterogeneous graphs by stacking multiple network layers introduces significant noise and can easily lead to oversmoothing. Consequently, Heterogeneous Graph Neural Networks (HGNNs) have emerged. DTInet¹¹ integrates multiple data sources to construct a heterogeneous network for DTI prediction. NeoDTI¹² also builds a heterogeneous network containing drug and protein information and employs network embedding techniques to learn low-dimensional representations of drugs and proteins, predicting interactions based on their proximity. EEG-DTI¹³ uses a heterogeneous GCN to learn drug and target representations in a heterogeneous network and employs an end-to-end framework for DTI prediction. SGCL¹⁴ uses a meta-path-based graph encoder with

attention mechanisms to learn drug and protein node representations from a heterogeneous network, constructing topological and semantic networks for DTI prediction. FuHLDR¹⁵ uses a graph convolutional network for lower-order representation learning and a meta-path-based strategy for higher-order relationships among drugs, proteins, and diseases. SHGCL-DTI¹⁶ introduces neighbor-view and meta-path-view encoders for node aggregation and uses graph contrastive learning to obtain node representations, enhancing DTI prediction performance and the framework's generalizability. LG-DTI¹⁷ employs a dual-view approach to DTI prediction, combining a local view, which extracts attribute features from biological information such as molecular structures and protein sequences, and a global view, which captures behavior features of drugs and proteins from the topological structure of a heterogeneous information network (HIN). MHGNN-DTI¹⁸ uses a meta-path-aggregated heterogeneous graph neural network to capture high-order associations between drug–target pairs in a biological heterogeneous graph for DTI prediction. MHTAN-DTI¹⁹ applies a meta-path-based hierarchical transformer and attention network, utilizing meta-path instance-level transformers, single-semantic attention, and multi-semantic attention on the heterogeneous bioinformatics network to generate low-dimensional vector representations of drugs and proteins for accurate DTI prediction.

Nevertheless, current heterogeneous graphs for DTI prediction typically include only drug, protein, disease, and side effect nodes. Despite the detailed information and rich semantics provided by these heterogeneous graphs, there is substantial potential to enhance prediction performance by integrating additional data sources.²⁰ The inclusion of multi-omics data, such as molecular attributes, protein interactions, and GO annotations, could provide a more comprehensive view of the biological network, thereby enriching the feature space

and improving the predictive power of the models. Furthermore, the majority of existing multiview approaches primarily utilize only two views, which limits the interaction and integration of information. The potential for multiview interactions is vast,²¹ as different views can complement each other by providing diverse perspectives on the same biological phenomena. However, the challenge lies in effectively integrating and fusing information from multiple views. Therefore, it is crucial to develop effective methods for multiview fusion that can balance the contributions from each view, preserve the unique information they provide, and enhance their collective predictive capability.

To address the aforementioned challenges, we propose a novel heterogeneous graph multiview contrastive learning model MVCL-DTI for predicting DTIs. First, we introduce two effective features for DTI prediction tasks: GO and substituents, which supplement the biological information on nodes and edges in the heterogeneous graph. While other methods integrate GO and substituents as part of the feature encoding for targets and drugs,²² our approach treats GO and substituents as distinct nodes within the heterogeneous graph structure. The model then learns representations from three different views of the heterogeneous graph, namely neighbor view, meta-path view and diffusion view, capturing multidimensional information for comprehensive feature representation. Using a self-attention mechanism, each view remains relatively independent, and contrastive learning maximizes the similarity of the same protein and drug node representations across the three views. Additionally, a multiview attention weighted fusion mechanism is employed to adaptively integrate the information from these distinct views, ensuring that the most relevant features from each view are emphasized. Finally, we reconstruct the original heterogeneous graph to obtain DTI prediction results. Experimental results demonstrate that our model outperforms baseline methods on classic data sets and maintains robust performance in challenging scenarios. Specifically, MVCL-DTI excels in both AUROC and AUPR metrics of benchmark under various positive-to-negative sample ratios. Furthermore, it demonstrates robust performance in experiments involving the selection of hard negative samples, masking known and redundant DTIs, and evaluating unique DTIs, showcasing its adaptability and effectiveness in diverse scenarios. The model's potential to discover new therapeutic targets was supported by literature-validated novel DTIs, and its competitive performance in real-world applications, such as COVID-19 drug studies, was highlighted. Feature visualization revealed the model's ability to capture rich, high-dimensional features, while computational property analysis identified key factors contributing to MVCL-DTI's superior performance, including effective feature representations, integration of GO and substituent node feature representations, and the importance of feature initialization strategy. Hyperparameter tuning during training was crucial for the model's success. Comprehensive experiments indicate that MVCL-DTI's robust performance is a result of deliberate design and careful training.

2. MATERIALS AND METHODS

The architecture of MVCL-DTI is illustrated in Figure 1. Our approach comprises four main components: First, we construct the heterogeneous graph with drug, protein, disease, side effect, substituents and GO nodes. Their interactions, associations, or similarities are depicted as various types of edges. Second, we learn representations of the heterogeneous graph from three distinct views, namely neighbor view encoder, meta-path view

encoder, diffusion view encoder, enabling the model to capture multidimensional information. The neighbor view encoder captures the 2-hop neighbor information for each node in the heterogeneous graph. The meta-path view encoder derives node features from meta-path-based neighbor nodes. The diffusion view encoder further captures higher-order neighbor information through random walks on the heterogeneous graph. Third, we employ contrastive learning with attention mechanism to derive node representations by effectively integrating information from different views. Additionally, a multiview attention weighted fusion module is utilized to further enhance the integration process, allowing the model to adaptively assign importance to each view for more robust and comprehensive feature representations. Finally, we reconstruct the original heterogeneous network using the node representations from the neighbor view, with the reconstructed DTI network serving as the prediction result.

2.1. Data Set. We have chosen Luo's data set¹¹ as the foundation to construct our own data set for the DTI task. Luo's data set is a classic data set used to construct heterogeneous networks for predicting DTIs. It comprises four types of nodes collected from public databases: drug nodes from the DrugBank database,²³ protein nodes from the HPRD database,²⁴ disease nodes from the Comparative Toxicogenomics Database,²⁵ and side effect nodes from the SIDER database.²⁶ The interactions between these nodes form eight types of relation networks: the "Drug–Side effect Association Network" obtained from the SIDER database, the "Protein–Disease Association Network" and "Drug–Disease Association Network" from the Comparative Toxicogenomics Database, the "Drug–Protein Interaction Network" and "Drug–Drug Interaction Network" from the DrugBank database, and the "Protein–Protein Interaction Network" was obtained from the HPRD database. Additionally, the "Drug Chemical Structure Similarity Network," where drug–drug similarity is calculated based on the chemical structure of the drug using the Tanimoto coefficient,²⁷ and the "Protein Sequence Similarity Network," where protein–protein similarity is calculated based on genomic sequences using the Smith–Waterman algorithm.²⁸ Overall, Luo's data set consists of 12,015 nodes and 1,894,854 edges.

Additionally, we added two additional node types: GO collected from Uniprot and drug substituents collected from the DrugBank database:

- **GO:** The Gene Ontology describes biological knowledge in terms of three aspects: Molecular function, Biological process, and Cellular component, often associated with gene and protein functions.
- **Substituents:** Drug substituents refer to functional groups in drug molecules where certain atoms or groups replace one or more hydrogen atoms. Substituents can significantly affect the physical, chemical, and biological properties of drug molecules, altering their activity, selectivity, and pharmacokinetic properties.

Meanwhile, two new edges were introduced: the "Protein–GO Association Network" from Uniprot and the "Drug–Substituents Association Network" from the DrugBank database. These additions enhance the data set's comprehensiveness and provide a more detailed representation of drug-related interactions. Our newly constructed data set now includes 22,225 nodes and 1,953,082 edges. Detailed statistics can be found in Table 1. We identified 1,923 DTIs within this heterogeneous graph and named this data set D1923. Positive

Table 1. Detailed D1923 Statistics

Node	Count	Edge	Count
Drug	708	Drug–Protein	1,923
Protein	1,512	Drug–Drug	10,036
Disease	5,603	Drug–Disease	199,214
Side-effect	4,192	Drug–Side-effect	80,164
GO	9,279	Drug–Substituents	14,896
Substituents	931	Protein–Protein	7,363
Total	22,225	Protein–Disease	1,596,745
		Protein–GO	42,741
		Total	1,953,082

samples for DTI prediction were selected from known DTIs, while negative samples were chosen from unknown DTIs. Therefore, the number of positive and negative samples in this heterogeneous graph is 1,923 and 1,068,573, respectively. It is important to note that the number of negative samples used in training can be randomly selected based on different experimental settings.

To capture diverse interactions, we defined the following meta-paths: drug–drug, drug–protein–drug, drug–disease–drug, drug–side effect–drug, drug–substituents–drug, protein–protein, protein–drug–protein, protein–disease–protein, and protein–GO–protein. Meta-paths consider not only direct relationships between nodes but also indirect relationships connected through multiple nodes and edges. For example, drug–protein–drug indicates that drugs are associated through the same protein, and drugs interacting with the same protein may share similar properties. Drug–disease–drug indicates that drugs are associated through treating the same disease, and drugs treating the same disease may have similar properties. Drug–side effect–drug indicates that drugs are associated through causing the same side effect, and drugs causing the same side effect may have similar properties. Thus, meta-paths can capture more complex graph structures and semantic information and are crucial for extracting meaningful relationships within the heterogeneous network.

2.2. Heterogeneous Network Construction. A heterogeneous graph can be represented as $G = (V, E)$, where V and E denote the sets of nodes and edges in the graph, respectively, while A and R denote the sets of node types and edge types, respectively. In a heterogeneous graph, multiple node and edge types exist, i.e., $|A| + |R| > 2$.

We extended Luo's data set¹¹ to construct our heterogeneous graph by introducing two new nodes: GO and substituents, and two corresponding new edges: drug–substituents and protein–GO associations. Therefore, in this work, $A = \{\text{drug, protein, disease, side effect, GO, substituents}\}$ and $R = \{\text{drug–drug interaction, drug–protein interaction, drug–disease association, drug–side-effect association, drug–substituents association, protein–protein interaction, protein–disease association, protein–GO association, drug–drug similarity, protein–protein similarity}\}$. For the eight interaction or association networks, each edge is assigned a weight q of 1. In the case of two similar networks, the weights q of the edges are based on the similarity score between drugs or the similarity score between proteins. We then normalize these similarity scores to fall within the range of 0–1, using them as the edge weights.

Meta-paths, consisting of alternating types of nodes and their relationships, are abstract paths used to describe the relationships between nodes and edges in a heterogeneous network. Meta-paths consider not only direct connections between nodes

but also indirect connections through different types of nodes and relationships, capturing high-order structural and semantic information in heterogeneous networks. We use $P = a_1 \xrightarrow{r_1} a_2 \xrightarrow{r_2} \dots \xrightarrow{r_{n-1}} a_n$ to represent a meta-path, where $a_i \in A$ and $r_i \in R$. In this paper, the neighbors of node v_i based on a meta-path are defined as nodes connected to v_i through the meta-path P_n .

2.3. Node Feature Initialization. For the initialization of drug node features, molecular representations are used to describe the molecules in a human-readable format, with SMILES (Simplified Molecular Input Line Entry System) being a common format.^{29,30} SMILES simplifies chemical structures with a strict syntax. Additionally, for a more comprehensive understanding of drug molecular structures, chemical fingerprints^{31,32} were established. After experimenting with various methods, we chose to use the MACCS fingerprint based on SMILES expressions³³ as the input for drug nodes. This expression defines 166 molecular descriptors. To extract the most important molecular descriptors, we employed Principal Component Analysis (PCA) to reduce the dimensions to 128.

For protein nodes, advances in pretrained models in the Natural Language Processing (NLP) domain have been successfully applied to learn hidden biological information from protein sequences. We utilized the ESM2 protein language model³⁴ to extract protein features from their sequences. Similar to the drug processing, we used PCA to reduce the dimensionality of protein node descriptors to 128.

For the remaining nodes: disease, side effect, GO, and substituents, we followed the approach of Yao et al.,¹⁶ and employed the node2vec³⁵ method to initialize these node features. Node2vec provides a mechanism to balance between breadth-first search (BFS) and depth-first search (DFS), which allows for better capturing of both local and global structures of the graph by learning node embeddings through random walks. Node2vec translates contextual relationships into features of drug–drug interaction, drug–disease association, drug–side-effect association, drug–substituents association, protein–protein interaction, protein–disease association, and protein–GO association. The output dimension is set to 128, consistent with the dimensions of drug and protein features.

2.4. Neighbor View Encoder. We utilize a Heterogeneous Graph Convolutional Network (HGCN)³⁶ to extract node features from the constructed heterogeneous graph. For each node, we first group its first-order neighbors according to edge types so that different types of edges are aggregated separately. When the HGCN has only one layer, nodes can only aggregate information from first-order neighbors. Stacking N layers of graph convolution allows nodes to aggregate information from N -order neighbors. The larger the N , the wider the aggregation range, but excessively large N can lead to gradient vanishing. Through experiments, we set N to 2. The aggregation process for each HGCN layer is defined as follows:

$$h_i^{(l+1)} = \frac{1}{1 + |R|} \left(\sum_{r \in R} \sum_{j \in N_i^r} \frac{q_{i,j,r}}{Q_{i,r}} \sigma(W_r^{(l)} h_j^{(l)} + b_r^{(l)}) + h_i^{(l)} \right) + h_i^{(l)} \quad (1)$$

where l indicates the layer number, N_i^r represents the set of neighbors of node v_i under edge type r , $|R|$ is the number of edge types, $Q_{i,r} = \sum_{j \in N_i^r} q_{i,j,r}$ represents the edge weight, $q_{i,j,r}$

and $Q_{i,r}$ are normalization constants, $W_r^{(l)}$ and $b_r^{(l)}$ are the weights and biases of the fully connected layers for edge type r , and σ is the activation function (ReLU). $h_i^{(l)}$ and $h_j^{(l)}$ denote the v_i node features and neighbor features, respectively. To address the potential over smoothing problem when stacking multiple layers, we introduce residual connections in the HGNC architecture. Specifically, the feature representation of each node in $l + 1$ layer is obtained by adding the transformed features from its neighbors to the original features from layer l . This residual design ensures that the model retains low-level information while learning higher-order interactions effectively. When $l = 0$, the initialized node features are used as input. Ultimately, we obtain the node features h^{ne} for the neighbor view.

2.5. Meta-Path View Encoder. Using predefined meta-paths, we employ a Graph Neural Network (GNN-FiLM)³⁷ to obtain node representations in the meta-path view. GNN-FiLM, based on Feature-wise Linear Modulation, generates different shifting (β) and scaling (γ) parameters for different types of nodes, dynamically adjusting the node feature representations based on each node's type:

$$\beta_{p_n,i}^{(l)}, \gamma_{p_n,i}^{(l)} = g(h_i^{(l)}; \theta_{g,p_n}) \quad (2)$$

Where g is a learnable function used to compute the parameters for affine transformation, θ_{g,p_n} is the parameter of the learnable function for meta-path P_n .

In GNN-FiLM, the target node's representation is applied to all incoming information through transformations, achieving feature modulation-based information transmission. The target node's representation is modulated based on its type using the FiLM parameters $\beta_{p_n,i}^{(l)}$ and $\gamma_{p_n,i}^{(l)}$:

$$h_{i,p_n}^{(l+1)} = \sigma \left(\sum_{j \in N_{i,p_n}^{p_n}} \gamma_{p_n,i}^{(l)} \odot (W^{(l)} h_j^{(l)}) + \beta_{p_n,i}^{(l)} \right) + h_{i,p_n}^{(l)} \quad (3)$$

Where $N_{i,p_n}^{p_n}$ is the set of neighbors of node v_i based on meta-path P_n , $W^{(l)}$ is weight matrix of the l -th layer. GNN-FiLM is particularly effective in handling heterogeneous graphs containing multiple types of nodes and edges, as opposed to standard GCNs, which do not differentiate between node types and treat all nodes uniformly.

GNN-FiLM allows us to capture deep semantic information within heterogeneous graphs, thus enhancing the model's comprehension of complex graph structures. Since different types of meta-paths encapsulate diverse semantic information, we use semantic-level attention to aggregate the results from different meta-paths. First, we compute the importance of each meta-path as follows:

$$W_{p_n} = \frac{1}{|V|} \sum_{i \in V} \alpha^T \cdot \sigma(W h_{i,p_n}^{(l)} + b) \quad (4)$$

Where W is weight matrix, b is the bias vector, and α is the semantic-level attention vector. After obtaining the importance of each meta-path, we normalize them using softmax to get the weight β_{p_n} of meta-path P_n :

$$\beta_{p_n} = \text{softmax}(W_{p_n}) = \frac{\exp(W_{p_n})}{\sum_{p \in P} \exp(W_p)} \quad (5)$$

Finally, we obtain the final node representation:

$$h_i^{(l)} = \sum_{n=1}^m \beta_{p_n} \cdot h_{i,p_n}^{(l)} + h_i^{(l-1)} \quad (6)$$

Where m represents the number of meta-paths. Similar to the neighbor view, by using a two-layer meta-path view encoder with residual connection and L2 regularization, we obtain the node representations for the meta-path view h^{mp} .

2.6. Diffusion View Encoder. To mitigate the over-smoothing effect caused by multilayer graph convolution operations, we initially apply a one-hop graph convolution operation, as eq 1.

After the graph convolution, we use the Random Walk with Restart (RWR)³⁸ strategy for further information propagation. Applying RWR on the heterogeneous graph enables the model to focus on higher-order neighbors without oversmoothing. In the random walk graph, nodes move through the graph in a series of random steps, typically determined by a probability distribution. This model helps study the process of information diffusion between nodes, capturing connections over a larger range. If the transition matrix of the heterogeneous network is A and the restart probability is ∂ , then RWR is defined as

$$A_{ppr} = \partial(I - (1 - \partial)A)^{-1} \quad (7)$$

where I is the identity matrix, and $A_{ppr}(u, v)$ represents the influence of node u on node v .

According to eq 7, we can spread node information over long distances to get the final node embedding vector:

$$h^{(l)} = \partial(I - (1 - \partial)A)^{-1} h^* + h^{(l-1)} \quad (8)$$

where h^* is the node embedding matrix obtained from the convolution operation. Similar to the previous two views, by using a two-layer diffusion view encoder with residual connection and L2 regularization, we obtain the node representations for the diffusion view h^{df} .

2.7. Contrastive Learning. First, we pass the node embedding matrices from the three views separately using a self-attention mechanism layer. By computing within the same view, the self-attention mechanism³⁹ allows different parts of the model to "attend" to each other, enhancing the model's representation capabilities. The self-attention mechanism helps the model better capture important features and relationships, improving the quality and accuracy of representations. We compute embeddings for neighbor view as an example:

$$h_{att}^{ne} = \text{Attention}(h^{ne}) \quad (9)$$

We employ a Multi-Layer Perceptron (MLP) with hidden layers to map h_{att}^{ne} , h_{att}^{mp} , and h_{att}^{df} into spaces suitable for calculating contrastive loss, resulting in z^{ne} , z^{mp} , and z^{df} . Next, we define pairs of the same drug or protein nodes across different views as positive pairs and other node pairs as negative pairs. For a node v_i , Pos_i represents the set of positive samples, while the remaining nodes form the negative sample set Neg_i . Using Info Noise Contrastive Estimation loss (InfoNCE),⁴⁰ we compute the contrastive loss between two views, such as the neighbor view and the meta-path view:

$$L_k^{NEtoMP} = \frac{1}{|V|} \sum_{i \in V_k} -\log \left(\sum_{j \in Pos_i} \text{softmax} \left(\frac{\text{sim}(z_i^{ne}, z_j^{mp})}{\tau} \right) \right) \quad (10)$$

$$L_k^{MPtoNE} = \frac{1}{|V|} \sum_{i \in V_k} -\log \left(\sum_{j \in Pos_i} \text{softmax} \left(\frac{\text{sim}(z_i^{mp}, z_j^{ne})}{\tau} \right) \right) \quad (11)$$

$$L^{ne-mp} = \sum_{k \in K} L_k^{NEtoMP} + L_k^{MPtoNE} \quad (12)$$

Where τ is the temperature coefficient, V_k is the set of nodes with type k , $k \in K = \{\text{drug}, \text{protein}\}$, and sim denotes the cosine similarity. We maximize the similarity between the representations of positive pairs, enabling nodes to learn representations from the three views.

The contrastive loss L^{ne-mp} , L^{ne-df} , and L^{mp-df} are computed similarly, providing representations used for subsequent DTI prediction:

$$Loss_{contrastive} = L^{ne-mp} + L^{ne-df} + L^{mp-df} \quad (13)$$

2.8. Multiview Attention Weighted Fusion. To further integrate the features extracted from different views, we introduced a multiview attention fusion module inspired by RGLDR.⁴¹ The learned representations h^{ne} , h^{mp} and h^{df} from the contrastive learning are passed into the attention fusion module, which assigns importance weights to each view dynamically and effectively aggregates their information. The procedure can be summarized as follows:

For each view $i \in \{ne, mp, df\}$, the attention score e^i is calculated as

$$e^i = W_{att}^T \cdot \tanh \left(W \cdot \frac{1}{|V|} \sum_{k \in I} h_k^i + b \right) \quad (14)$$

Where h_k^i denotes the feature vector of node k in view i , W and W_{att} are learnable weight matrices, and b is the bias term. This step captures high-level interactions across views.

The attention coefficients β_i , representing the relative importance of each view, are then normalized using the softmax function:

$$\beta_i = \frac{\exp(e^i)}{\sum_{j \in \{ne, mp, df\}} \exp(e^j)} \quad (15)$$

Finally, we compute the fused representation H_k for each node type k as the weighted sum of view-specific representations:

$$H_k = \sum_{i \in \{ne, mp, df\}} \beta_i \cdot h_k^i \quad (16)$$

2.9. Reconstruct Heterogeneous Network for DTI Prediction. After feature aggregation and information propagation, we reconstruct the heterogeneous network using the obtained representations of six types of nodes and employ the DistMult⁴² method as the scoring mechanism. Specifically, for the ten types of edges in the original heterogeneous graph, we use DistMult to reconstruct them. The reconstructed DTI network is the prediction result, defined as

$$N_{reconstruction}^{dti} = H_{drug}^T M_{dti} H_{protein} \quad (17)$$

Where H_{drug}^T and $H_{protein}$ are the drug and protein node feature matrices obtained from the previous steps, and M_{dti} is a learnable edge-type specific diagonal matrix. The reconstruction process for other types of edges is similar to that of the DTI network.

The loss function is calculated as follows:

$$Loss_{reconstruction} = \sum_{r \in R} (Q^r)^2, \quad Q^r = N_{original}^r - N_{reconstruction}^r \quad (18)$$

Where $N_{original}^r$ and $N_{reconstruction}^r$ are the original and reconstructed heterogeneous network of edge type r . Q is a matrix storing the difference between the predicted and true values.

Finally, adding the L2 regularization term, we obtain the final objective function:

$$\begin{aligned} Loss &= Loss_{reconstruction} + \lambda_1 Loss_{contrastive} + \lambda_2 Loss_{L_2} \\ &= Loss_{reconstruction} + \lambda_1 Loss_{contrastive} + \lambda_2 \sum_w w^2 \end{aligned} \quad (19)$$

Where λ_1 and λ_2 are hyperparameters, w represents weights in the model. The goal of this paper's training process is to minimize this objective function.

3. EXPERIMENTAL EVALUATION AND DISCUSSION

3.1. Parameter Settings. For our model, we set the number of network layers per view encoder to 2, with hidden layer dimensions of 2,048. The learning rate was set to 1e-3, the temperature parameter τ to 0.5, and the dropout rate to 0.2. We used the Adam optimizer with an early stopping mechanism, running for 5,000 iterations.

3.2. Evaluation Metrics. DTI prediction can be viewed as a binary classification task, so we used AUROC (Area Under the Receiver Operating Characteristic Curve) and AUPR (Area Under the Precision Recall Curve) as evaluation metrics. AUROC is a global metric for evaluating classifier performance. It measures the classifier's performance across different decision thresholds by plotting the relationship between true positive rate (TPR, also known as recall) and false positive rate (FPR). AUROC values range from 0.5 to 1, with higher values indicating better model performance. In extremely imbalanced data sets, the false positive rate may be very low, making the AUROC value potentially misleading. Even if the false positive rate is very low, the AUROC may still perform well, but the model may not effectively distinguish the minority class in practice. AUPR, on the other hand, evaluates the relationship between precision and recall at different thresholds, with values ranging from 0 to 1. AUPR is particularly suitable for evaluating model performance on imbalanced data sets because it focuses more on the prediction of the minority class. To ensure balanced experimental results, we employed a 10-fold cross-validation setup across the data set. The results of other methods are taken from their respective papers.

3.3. Baselines. When selecting comparison methods, we prioritized representative and state-of-the-art approaches, including SHGCL-DTI,¹⁶ to validate the competitiveness of our model. Additionally, we incorporated two classical machine learning methods, NetLapRLS⁶ and DTInet,¹¹ which are widely recognized and often employed as benchmarks in the field. The remaining methods are advanced GNN-based approaches: NeoDTI¹² and EEG-DTI.¹³ Ultimately, to evaluate the performance of our model, we conducted comparisons with five baseline methods using Luo's data set:

- **NetLapRLS:** NetLapRLS employs a semi-supervised learning algorithm using Laplacian Regularized Least Squares, incorporating both similarity and interaction information into the prediction framework.

Table 2. Results of Different Ratios of Positive and Negative Samples

Ratio	1:1		1:10		1: all	
	AUROC	AUPR	AUROC	AUPR	AUROC	AUPR
NetLapRLS	0.9412	0.9485	0.9073	0.7406	0.8891	0.4081
DTInet	0.9302	0.9538	0.9150	0.8184	0.9072	0.4402
NeoDTI	0.9455	0.9555	0.9452	0.8495	0.9153	0.5808
EEG-DTI	0.9551	0.9643	0.9554	0.8747	0.9532	0.1697
SHGCL-DTI	0.9510	0.9621	0.9582	0.8851	0.9234	0.6098
MVCL-DTI	0.9604	0.9674	0.9677	0.9024	0.9449	0.6728

*Bold values in the table denote the highest performance.

• DTInet: DTInet uses unsupervised learning to derive low-dimensional feature representations of drugs and targets from heterogeneous network data. It ranks candidate targets based on the geometric proximity of their feature vectors to the drug's vector, predicting new drug–target interactions.

• NeoDTI: NeoDTI applies GCNs to heterogeneous networks to learn drug and target representations. It optimizes feature extraction and DTI prediction in an end-to-end manner, predicting interactions by reconstructing the original network.

• EEG-DTI: EEG-DTI is an end-to-end heterogeneous graph representation learning method. It uses GCNs to generate feature representations for nodes within the heterogeneous network. The final step involves applying an inner product method to predict DTIs.

• SHGCL-DTI: SHGCL-DTI constructs a neighbor view encoder and a meta-path view encoder for node aggregation. This method utilizes graph contrastive learning to obtain robust node representations. By contrasting different views, SHGCL-DTI enhances the performance and generalization ability of the DTI prediction framework.

3.4. Prediction Accuracy of MVCL-DTI on Benchmark.

3.4.1. The Impact of Positive-to-Negative Sample Ratios on Model Performance. In real-world scenarios, confirmed drug–target interactions are sparse, resulting in a significantly larger number of negative samples compared to positive ones. We conducted experiments with different ratios of positive and negative samples to investigate the impact of varying sample ratios on the experimental results and to validate the model's generalization capability. The experimental results are presented in Table 2.

First, we constructed a data set with a balanced 1:1 ratio of positive and negative samples, where all known DTIs served as positive samples, and an equal number of negative samples were randomly selected from unknown DTIs. Our model outperformed all baseline methods, achieving a 1.38% higher AUROC and a 0.93% higher AUPR than the second-best model. This superior performance in a balanced data set setting can be attributed not only to the addition of two new nodes that enrich the biological information of drugs and proteins but also to our three-view contrastive learning framework, which captures more effective embeddings to enhance prediction accuracy.

Extending the imbalance, we established a data set with a 1:10 positive-to-negative ratio, maintaining all known DTIs as positive samples and randomly selecting ten times more unknown DTIs as negative samples. Again, our model demonstrated superiority, surpassing all baselines with AUROC and AUPR scores 0.95% and 1.73% higher, respectively, than the second-best method. Furthermore, we have also evaluated the performance of our model using additional metrics, including accuracy, precision, recall, F1 score, and MCC, under positive-to-negative sample ratios of 1:1

and 1:10. The results are provided in Tables S1 and S2, demonstrating the robustness of our model across different evaluation criteria.

To further exacerbate the imbalance, we established a data set where all known DTIs comprised the positive samples, and all unknown DTIs served as negative samples. Our model achieved the best performance in terms of AUPR, surpassing the second-best method by a significant 6.3%. In terms of AUROC, our model achieved the second-best result, 0.83% lower than EEG-DTI. However, for highly imbalanced data, AUROC may be overly lenient toward false positive rates, thereby overestimating the actual performance of the model. In practical applications, we are usually more concerned with whether the model can correctly identify the minority class, thus AUPR better reflects this need.

As shown in Figure S1, the main metric AUPR gradually decreases as the ratio of negative samples increases. However, the decrease in our method is relatively small, with reductions of 0.069 and 0.2296, as demonstrated in Table S3. Therefore, the significant improvement in AUPR demonstrates the superior predictive capability of our model, especially in real-world scenarios with sparsely labeled DTI.

To further enhance the reliability of the advantages of MVCL-DTI, we conducted comprehensive statistical analysis to assess the significance of performance differences among the six compared methods. Using the results from 10-fold cross-validation for AUROC and AUPR under positive-to-negative sample ratios of 1:1 and 1:10, we applied the Friedman test, Friedman test, Nemenyi Test, and common statistical indicators including the Mean and Standard Deviation. The Friedman test, a nonparametric method designed to compare multiple related samples. This analysis revealed significant overall differences among the six methods, with average p-values of 0.0029248 for AUROC and 0.0014899 for AUPR. To pinpoint specific pairwise differences, posthoc Nemenyi tests were performed. These tests confirmed that our model significantly outperformed several baseline methods. Detailed pairwise p-values and adjusted p-values are presented in Tables S4 and S5. Additionally, we reported the mean, standard deviation (Std), and 95% confidence intervals (CI) for AUROC and AUPR across the 10 folds for each method as in Tables S6 and S7. Collectively, these metrics demonstrate the superior stability and effectiveness of MVCL-DTI, highlighting its advantage in handling different sample ratios and outperforming the other methods in terms of both statistical significance and practical performance.

3.4.2. The Effect of Random Masking Known DTIs at Various Ratios. To further analyze our model's ability to predict unknown DTIs, we randomly masked different ratios of known DTIs. We used the masked DTIs as positive samples in the testing set and the unmasked DTIs as positive samples in the

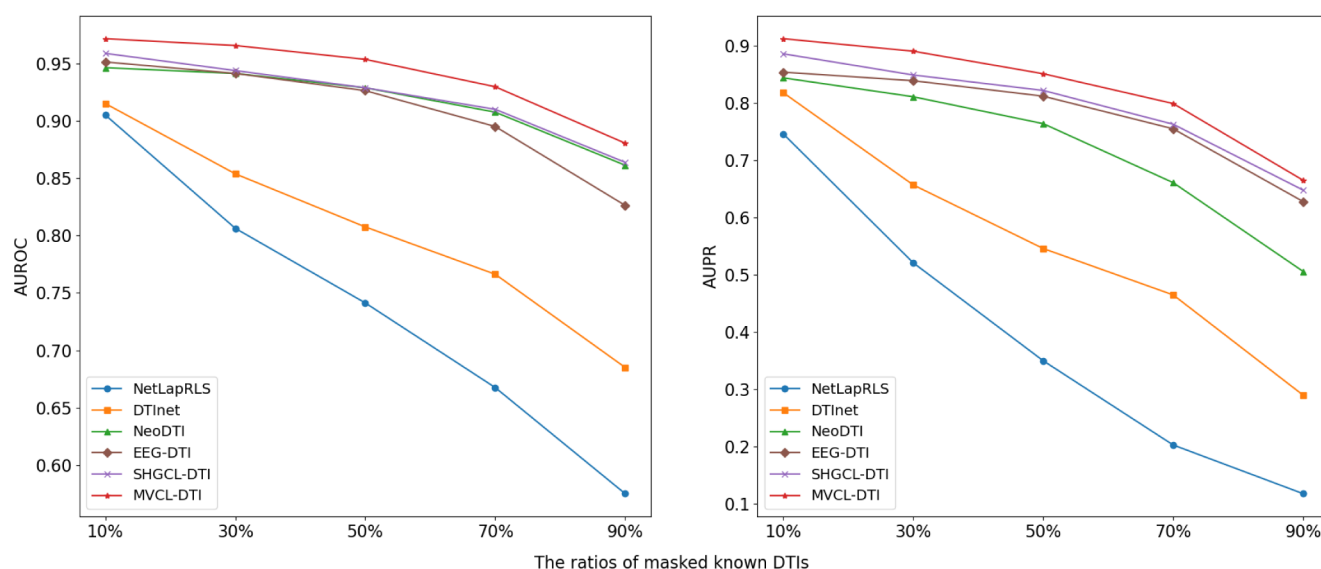


Figure 2. Comparative analysis of MVCL-DTI and baseline models on testing sets with varied masking ratios of known DTIs. The masked DTIs were used as positive samples in the testing set, while the unmasked DTIs served as positive samples in the training set. Negative samples in both the testing and training sets were randomly selected from unknown DTIs, maintaining a positive-to-negative sample ratio of 1:10.

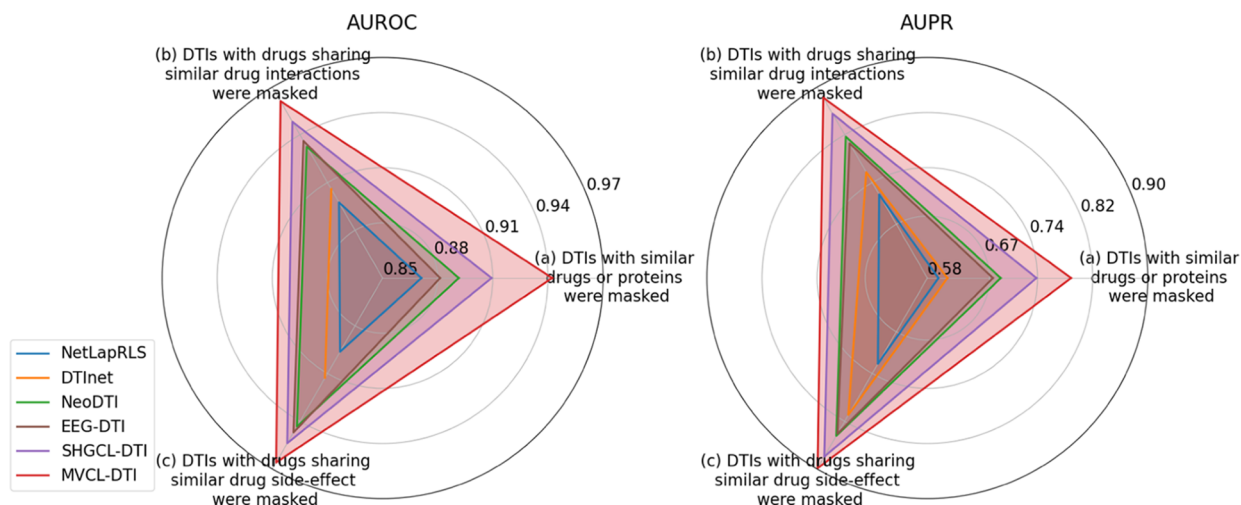


Figure 3. Comparative analysis of MVCL-DTI and baseline models on nonredundant DTIs: (a) masking DTIs involving drugs or proteins with high similarity, (b) masking DTIs with drugs sharing similar drug interactions, and (c) masking DTIs associated with drugs that have similar side effects. Throughout these three experiments, the ratio of positive-to-negative samples was consistently preserved at 1:10.

training set. Negative samples in both the testing and training sets were randomly selected from unknown DTIs, with a positive-to-negative sample ratio of 1:10. The masking ratios were 10%, 30%, 50%, 70%, and 90%. As shown in Figure 2, our model outperforms the second-best method across all masking ratios. Our model's AUROC and AUPR are higher than the second-best method at the 90% masking ratio by 1.67% and 1.71% respectively. For the other masking ratios of 10%, 30%, 50%, and 70%, our model significantly outperforms the second-best method, with AUROC improvements of 1.28%, 2.18%, 2.47%, and 1.97%, and AUPR improvements of 2.7%, 4.15%, 2.92%, and 3.62%, respectively. Additionally, it is evident that the prediction accuracy gradually decreases with an increasing ratio of masking. This indicates that the generalization ability of the model declines as the prior knowledge in the training set decreases. However, our method consistently maintains the best performance. These results indicate that our model exhibits strong generalization in predicting unknown DTIs.

3.5. Analysis of the Generalization Capability of MVCL-DTI. 3.5.1. Model Performance on Unknown DTIs.

The initial data set may contain redundant DTIs—such as homologous proteins or similar drugs, which raises concerns that the model's strong predictive performance might stem from interactions that are inherently easy to predict. To bolster the validation of MVCL-DTI's overall robustness, we executed two additional experiments, specifically designed as hard targets tests. These include validations on nonredundant DTIs and unique DTIs.

By masking these redundant DTIs, we can achieve more realistic results. We conducted the following additional experiments using the same data set processing method as Luo: (a) Masking DTIs with similar drugs (i.e., drug chemical structure similarities >0.6) or similar proteins (i.e., protein sequence similarities >40%); (b) Masking DTIs with drugs sharing similar drug interactions (i.e., Jaccard similarities >0.6); (c) Removing DTIs with drugs sharing similar side effects (i.e.,

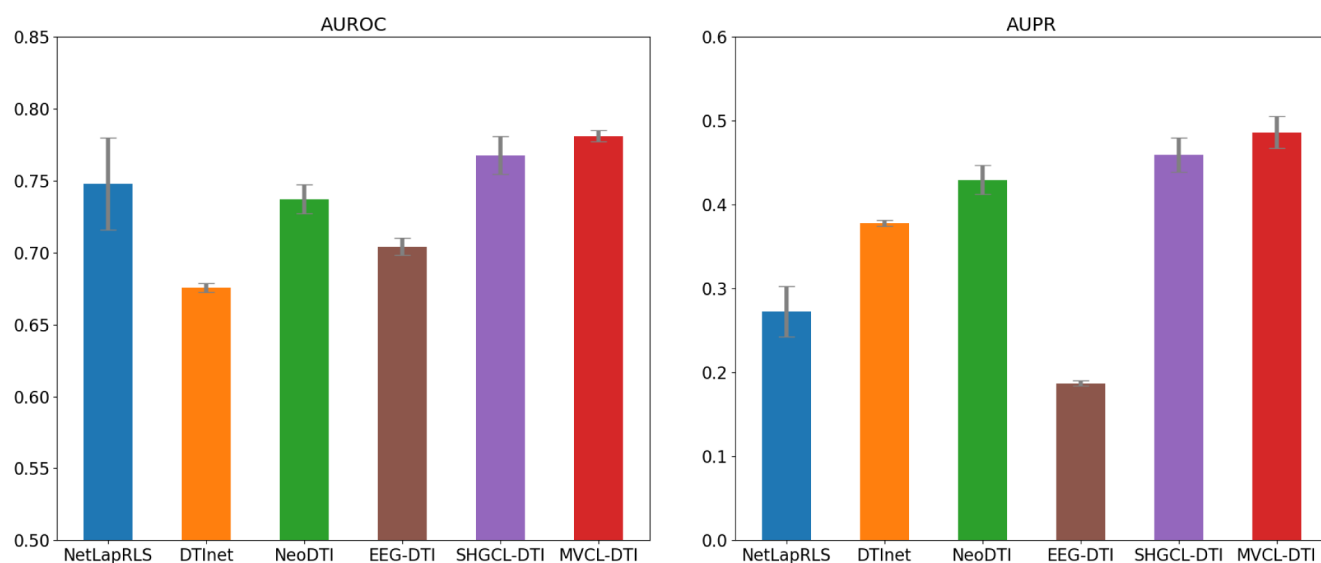


Figure 4. Comparative analysis of MVCL-DTI and baseline models trained on nonunique DTIs and tested on unique DTIs. Unique DTIs are characterized by interactions where a drug or target is exclusively connected to one corresponding target or drug. For the training and testing sets, the ratio of positive-to-negative samples was maintained at 1:10.

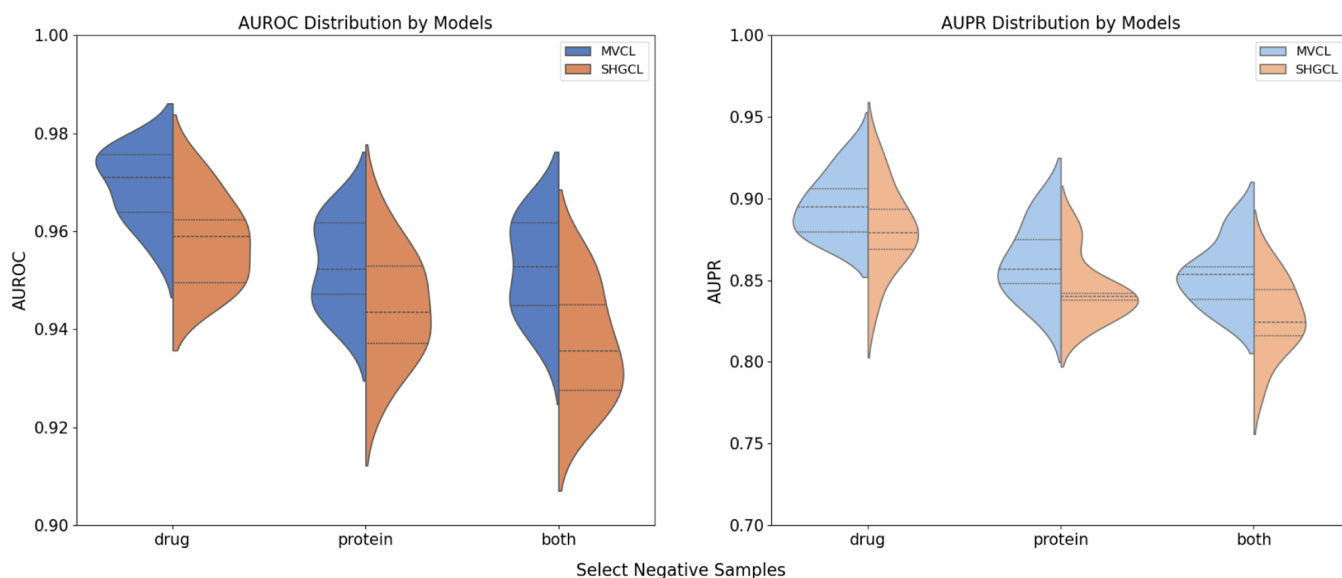


Figure 5. Comparative analysis of MVCL-DTI and SHGCL under three negative sample selection strategies. AUROC (left) and AUPR (right) distributions for MVCL-DTI and SHGCL under drug similarity-based, protein similarity-based, and combined similarity-based sampling scenarios. For the training and testing sets, the ratio of positive-to-negative samples was maintained at 1:10.

Jaccard similarities >0.6). The positive and negative ratio of the three experiments was 1:10. From the experimental results in Figure 3, our model still outperforms all baseline methods in these experiments, with our method's accuracy improving by an average of 1.94% and 3.22% over the second-best method across three data sets, confirming the stability and generalization of our model.

3.5.2. Model Performance on Unique DTIs. To further increase the difficulty of prediction, we consider DTIs where a drug (target) has only one interacting target (drug), which we refer to as unique DTIs. In binary classification tasks, if a data set contains multiple unique DTIs, traditional cross-validation may not be an appropriate method for evaluating predictive performance, as it may leverage biases toward those unique drugs and targets to improve performance. Therefore, we trained all methods on nonunique DTIs and tested them on

unique DTIs, ensuring isolation between the training and testing sets. The ratio of positive and negative samples in both the training and testing sets was 1:10. As shown in Figure 4, our model outperforms all baseline methods, with improvements of 1.34% in AUROC and 2.66% in AUPR over the second-best method. This demonstrates our model's stronger generalization capability and robustness in predicting new DTIs without extensive prior knowledge of known DTIs.

3.5.3. Model Performance on Hard Negative DTIs. In the initial design of our experiments, we employed a simple random sampling strategy for generating negative samples to ensure consistency and comparability with existing methods. However, this approach may lead to the inclusion of low-quality negative samples, potentially limiting the model's ability to learn effective feature representations. To address this issue, we conducted additional experiments using a more sophisticated negative

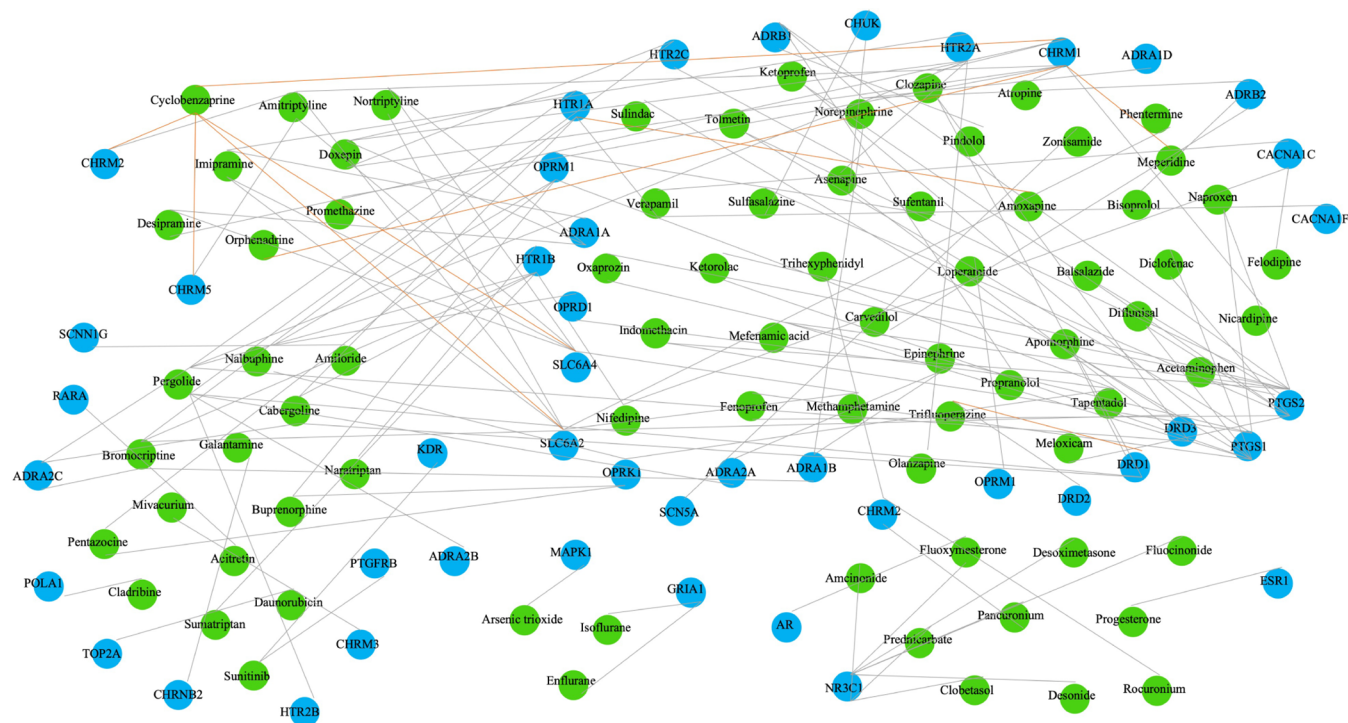


Figure 6. Network visualization of the top 150 DTIs predicted by MVCL-DTI model. Targets and drugs are depicted as blue circles and green circles, respectively. Gray edges indicate known drug–target interactions, while orange edges represent newly predicted interactions by MVCL-DTI.

sampling strategy that considers structural similarity, aiming to evaluate the impact of higher-quality negative samples on model performance. Specifically, we designed three different negative sampling scenarios: (1) Drug similarity-based sampling: Negative samples were selected from the data set where the chemical structure similarity between the drugs in the negative samples and the drugs in the positive samples exceeded 0.6. These samples are structurally similar to the positive samples, resulting in 864,453 negative DTIs. (2) Protein similarity-based sampling: Negative samples were chosen from the data set where the sequence similarity between the proteins in the negative samples and the proteins in the positive samples exceeded 40%. This yielded 347,121 negative DTIs, representing protein pairs that are sequence-similar but do not interact with the given drugs. (3) Combined drug and protein similarity-based sampling: Negative samples were selected where the drug chemical structure similarity exceeded 0.6 and the protein sequence similarity exceeded 40%, resulting in 280,566 negative DTIs. These samples represent high-confidence negative pairs where both drugs and proteins are structurally or sequentially similar to their counterparts in positive samples yet are biologically unrelated.

For each scenario, the number of positive samples in the data set was fixed at 1,923, and maintaining a 1:10 positive-to-negative sample ratio, and 10-fold cross-validation was performed. We evaluated the performance of our model against the SHGCL method under these conditions, using AUROC and AUPR as the primary metrics. The results were visualized in Figure 5. The violin plots depict the distribution of AUROC and AUPR scores for both methods. Across all scenarios, our model demonstrated consistently higher performance compared to SHGCL, with narrower distributions and higher median values for both metrics. This improvement underscores the robustness and adaptability of our model when working with higher-quality negative samples. These findings not only highlight the

importance of selecting meaningful negative samples but also validate the effectiveness of our approach in various settings.

3.6. Novel DTIs Identified by MVCL-DTI. Figure 6 presents the visualization of the network graph for the top 150 DTIs with the highest probability values predicted by MVCL-DTI. Targets and drugs are shown in blue circles and green circles, respectively. Known drug–target interactions are marked by gray edges, while the new predicted interactions are shown by orange edges. We grouped the 150 DTIs into four clusters based on drug similarity. It is evident that some clusters exhibit complex interactions, suggesting direct or indirect relationships among them. This further corroborates the idea that similar drugs may share similar interaction patterns. Moreover, proteins prefixed with DR are clustered together, which are genes or proteins related to the Recombinant Dopamine (DR) receptor. This indicates that drugs with similar therapeutic functions might be interconnected, reflecting the generalization ability of our method to uncover these indirect relationships.

Among the top 150 predicted DTIs, there are 9 newly predicted DTIs, of which 6 are supported by literatures, as detailed in Table 3. The strong correlation between our predictions and the existing literature indicate that the

Table 3. 19 Predicted DTIs with Literature Support in the Top 22 Newly Predicted DTIs

Drug name	Protein name	Support
Meperidine	CHRM1	drugbank
Amoxapine	HTR1A	drugbank
Orphenadrine	CHRM1	pubchem
Cyclobenzaprine	SLC6A4	drugbank
Cyclobenzaprine	SLC6A2	drugbank
Trifluoperazine	DRD1	pubchem

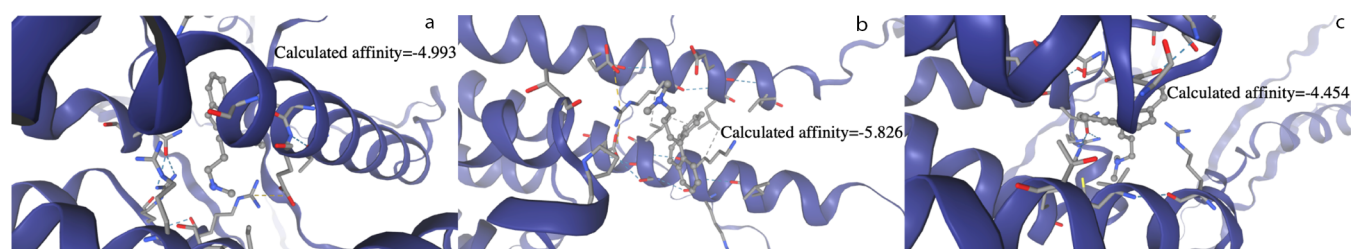


Figure 7. Computational docking validations of the 3 new predicted DTIs: (a) cyclobenzaprine-CHRM5, (b) cyclobenzaprine-CHRM1, and (c) cyclobenzaprine-CHRM2. All docking results obtained low docking scores, showing strong binding affinity.

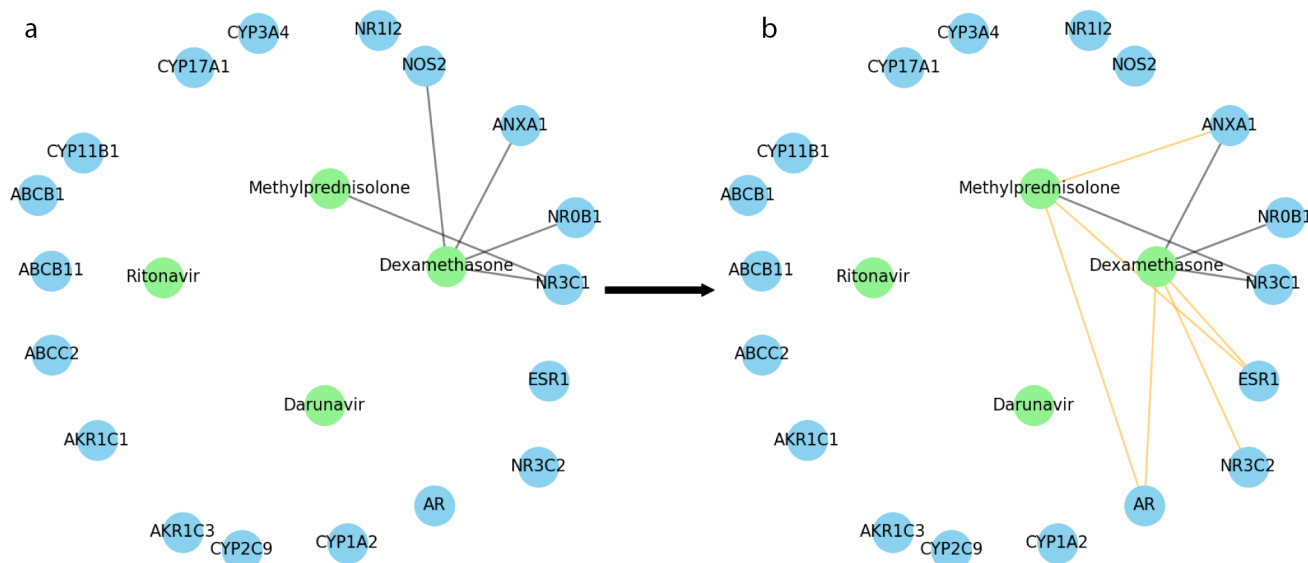


Figure 8. Drug–target interaction networks of 4 drugs interacting with targets. (a) The network derived from existing data set D1923 and (b) the network predicted by MVCL-DTI.

interactions predicted by our model are consistent with established scientific knowledge, thereby reinforcing the model's reliability.

For the 3 newly predicted DTIs that lack literature support—Cyclobenzaprine-CHRM5, Cyclobenzaprine-CHRM1, and Cyclobenzaprine-CHRM2, we performed molecular docking predictions using the online molecular docking prediction Web site SwissDock⁴³ and selected AutoDock Vina for the docking predictions. AutoDock Vina, an open-source and free tool, improves the accuracy of binding mode predictions and accelerates search speed using a simpler scoring function. We predicted the binding modes and docking scores for each DTI pair. The binding modes are shown in Figure 7, and the docking scores are -4.933 , -5.826 , and -5.454 , respectively. The docking scores from AutoDock Vina represent the calculated affinity (kcal/mol), where a lower value indicates stronger binding affinity. There is a general rule of thumb: calculated affinity > -4 kcal/mol indicates very weak or no binding; -7 kcal/mol $<$ calculated affinity ≤ -4 kcal/mol indicates moderate binding; calculated affinity ≤ -7 kcal/mol indicates strong binding. Therefore, the three DTIs we predicted without literature support fall into the moderate binding category according to the docking scores. Overall, the docking results of the newly predicted DTIs by our model may provide important clues for understanding the structural basis of these predictions, thereby aiding in the elucidation of the potential molecular mechanisms of drug action.

3.7. Validation MVCL-DTI Predictions on Targets of Drugs for COVID-19.

To further validate the effectiveness of our model in identifying potential DTIs, we conducted case studies on the following four drugs: Dexamethasone, Methylprednisolone, Ritonavir, and Darunavir. These drugs have been prominently used in the treatment of COVID-19. The COVID-19 pandemic, caused by the SARS-CoV-2 virus, has necessitated urgent therapeutic interventions. Identifying effective drugs and their interactions with viral and host proteins is crucial for developing treatments. Dexamethasone and Methylprednisolone are corticosteroids used to reduce inflammation and immune response, while Ritonavir and Darunavir are protease inhibitors commonly used in antiviral therapy, including for HIV and potentially for SARS-CoV-2.

We retrieved target proteins interacting with these drugs from the DrugBank and PubChem databases.⁴⁴ We then filtered out the target that exists in our data set, involving a total of 32 DTIs. These target proteins interact with the four COVID-19-related drugs, with some interactions already present in the heterogeneous graph derived from D1923, while others were newly identified. Additionally, there were some interactions that we did not predict. Detailed information is shown in Table S8. After training our model, we found that it could predict 6 potential drug–target interactions, and the changes are shown in Figure 8. The yellow lines in the figure represent newly predicted drug–target interactions. For Dexamethasone, out of 14 DTIs, our model predicted 6, whereas the SHGCL model predicted 4. For Methylprednisolone, our method predicted 1 more interaction

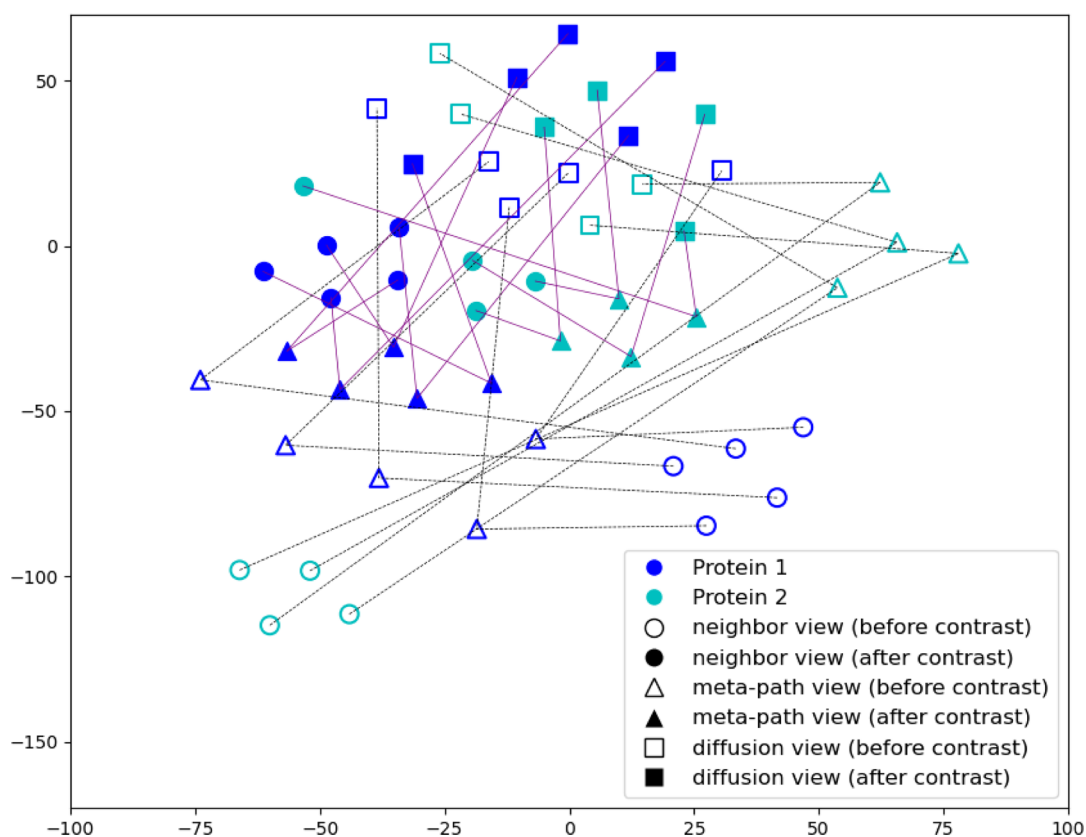


Figure 9. Visual analysis of protein features evolution across pre- and post-contrastive learning phases. Blue represents proteins1, which includes CHRM1, CHRM2, CHRM3, CHRM4, and CHRM5; cyan represents proteins2, which includes OPRD1, OPRK1, OPRM1, and OPRL1. Hollow and solid circles indicate the neighbor view before and after contrastive learning, respectively; hollow and solid triangles represent the meta-path view before and after contrastive learning; hollow and solid squares correspond to the diffusion view before and after contrastive learning. Black dashed lines and purple solid lines connect the same protein across the three views, before and after contrastive learning, respectively.

than the SHGCL model. This indicates that our model can effectively predict potential DTIs related with COVID-19, and highlights the practical utility of our model.

Furthermore, we observed that the degrees of the two drug nodes (Ritonavir and Darunavir) for which DTIs were not predicted were relatively small in the original heterogeneous graph. For example, the average degree of the two drug nodes (Dexamethasone and Methylprednisolone) with predicted DTIs was 1,280, while the average degree of these two drugs without predicted DTIs was 598, indicating that there was less prior knowledge about them, which impacted the learning performance of MVCL-DTI. If more relevant relational information is added, it is expected that more DTIs could be identified. Identifying more potential DTIs successfully could accelerate the repurposing of these drugs for COVID-19 treatment, potentially leading to quicker therapeutic interventions.

3.8. Visualization of Semantic Features Derived from MVCL-DTI. **3.8.1. Impact of Contrastive Learning on Protein and Drug Feature Representations.** To demonstrate the impact of the contrastive learning module, we selected two classes of proteins: proteins1 includes CHRM1, CHRM2, CHRM3, CHRM4, CHRM5, and proteins2 includes OPRD1, OPRK1, OPRM1, OPRL1. We used t-SNE⁴⁵ to visualize the representations of these nodes before and after contrastive learning, denoted by hollow and solid points, respectively. Different shapes represent different views (hollow and solid circles represent the neighbor view before and after contrastive

learning; hollow and solid triangles represent the meta-path view before and after contrastive learning; hollow and solid squares represent the diffusion view before and after contrastive learning), and different colors denote different classes (blue represents proteins1; cyan represents proteins2). The same protein across the three views is connected by lines, with black dashed lines connecting hollow points and purple solid lines connecting solid points. As shown in Figure 9, before contrastive learning, the representations of proteins within the same view are close to each other, such as the hollow triangles, hollow circles, and hollow squares clustered together. After contrastive learning, the proximity of the three views' protein representations indicates effective integration of the three views' information, as evidenced by the clustering of solid points. Moreover, the purple solid lines are noticeably shorter than the black dashed lines, indicating that the representations of the same protein across different views are closer, aligning with the objectives of contrastive learning.

The feature maps of the two classes of drugs (drugs1 includes Methylprednisolone, Fluoxymesterone, Spironolactone; drugs2 includes Triazolam, Midazolam, Alprazolam) before and after contrastive learning can be found in Figure S2, where it can be observed that the characteristic features of drugs are less pronounced than those of proteins. This is because the feature representations of both proteins and drugs are based on the information from the heterogeneous graph structure. In the heterogeneous graph, the average degree of each protein node is 1,090, while the average degree for each drug node is 432,

indicating that protein nodes contain more information. Given that the heterogeneous network related to proteins is more complex, the model is better able to learn representative feature representations. Therefore, if more drug-related interaction information is incorporated, the prediction accuracy of the model may be further enhanced.

3.8.2. Attention Weights in Feature Fusion. To further understand how the model integrates information from different node types and views, we visualized the attention weights learned during feature fusion in the diffusion view. These weights provide insights into the contribution of various relationships and views to the final feature representations. As shown in Figure S3, the drug–protein and protein–disease matrices contribute most significantly to the prediction performance for drug and protein nodes, respectively. This can be attributed to biological reasons: Drug–protein interactions form the core of DTI tasks as they represent the direct biochemical relationships between drugs and their target proteins; Protein–disease associations are critical for capturing the functional relevance of proteins within the biological network. Proteins often serve as biomarkers or therapeutic targets for specific diseases, and their connections to disease nodes provide contextual information about their role in pathological processes.

In addition, we visualized the attention weights across the three views (neighbor, meta-path, and diffusion views) during multiview attention weighted fusion, as shown in Figure S4. It is observed that the neighbor view contributes the most. This result can be explained from a model perspective: the neighbor view directly captures the first-order proximity in the graph, representing immediate connections between nodes. In a heterogeneous graph, these local connections often encode the most reliable and interpretable information, such as direct interactions between drugs and proteins or proteins and diseases. The model naturally prioritizes these relationships because they are less prone to noise compared to higher-order or inferred relationships. Meta-path and diffusion views, while valuable for capturing higher-order and global patterns, may dilute the direct signals critical for DTI tasks, explaining their comparatively lower weights.

3.9. Discussions on Computational Properties of MVCL-DTI. In this section, we present investigations into the computational properties of our model. We wish to justify that the results we presented in the previous sections are not coincident but contributed by intentional design and training of MVCL-DTI computational components.

3.9.1. Identifying Critical Features for DTI. It is not difficult to agree that representation features of protein and drug are necessary for DTI task. We found that ESM2 and MACCS are the best representations for MVCL-DTI. We also found that including GO and substituents node features can greatly enhance the performance of MVCL-DTI. Furthermore, we showed a proper initialization strategy for features trained by our MVCL-DIT model.

Initially, we delved into a variety of approaches for initializing the features of protein and drug nodes, as illustrated in Table 4. For proteins, we experimented with using protein language models such as ProtBERT⁴⁶ and ESM2³⁴ to extract protein sequence features. The results demonstrated that the features extracted using the ESM2 protein language model outperformed the baseline method on the DTI task. For drugs, we tried representing the SMILES strings as pharmacophore fingerprints,⁴⁷ ECFP,⁴⁸ Mordred descriptors,⁴⁹ and features extracted

Table 4. Comparative Analysis of MVCL-DTI Performance Utilizing Different Features of Protein and Drug Nodes on D1923 with a 1:10 Positive-to-Negative Sample Ratio

Feature type		AUROC	AUPR
protein	Protein-420 descriptors	0.9558	0.8814
	Protein-ProtBERT	0.9624	0.8897
	Protein-ESM2	0.9625	0.8900
drug	Drug-MACCS	0.9558	0.8814
	Drug-phar	0.9514	0.8751
	Drug-ecfp	0.9534	0.8716
	Drug-Mordred	0.9548	0.8739
	Drug-LLM	0.9516	0.8793

*Bold values in the table denote the highest performance.

from drug summaries using large language models. The results indicated that the baseline method remained the best. Therefore, we used ESM2 to extract protein features and represented drug SMILES sequences as MACCS fingerprints for drug features.

Then, subnetwork ablation experiments were conducted by masking different subnetworks related to drug or protein information to evaluate their contributions to the overall prediction performance. As seen in Figure 10, the model's performance was the worst when all heterogeneous networks were masked, leaving only the drug–protein interaction network. Next, masking the newly proposed GO and substituents subnetworks resulted in the second most significant performance drop, with AUROC and AUPR decreasing by 0.52% and 0.55%, respectively. This indicates that the novel proposed GO and substituents nodes, along with their corresponding new edges, provide more effective information for the DTI task when compared with traditional subnetworks like disease, side effect, DDI, and PPI networks, thereby affirming the advantage of our newly proposed nodes.

Furthermore, building upon the foundation of feature engineering, the initialization strategies for MVCL-DTI node features warrant exploration. Consequently, within our model, we have experimented with different node feature initialization strategies to assess their efficacy on DTI prediction:

- Random Strategy: Features for the four types of nodes were randomly initialized with a dimension of 128.
- Luo's Strategy:¹¹ DTINet applied diffusion component analysis (DCA) to learn the low-dimensional vector representations of the drug and target features that capture the intrinsic topological properties of a heterogeneous network. We directly used the publicly available node initialization features from Luo. For the remaining two node features, we used a random initialization strategy to obtain.
- Yao's Strategy (Baseline):¹⁶ The 166-bit MACCS molecular fingerprints were used as initial features for drugs, and 420 descriptors were constructed by extracting the relative ratio of amino acids and dipeptides from protein sequences. These features were then reduced to 128 dimensions using PCA.
- Our Strategy: Similar to the baseline, we used MACCS fingerprints with a dimension of 128 for drug initial features. For proteins, we extracted 1,280-dimensional sequence features using the ESM2 language model, which were then reduced to 128 dimensions using PCA. Additionally, we treated GO and substituents as new features, initializing their features using node2vec with a

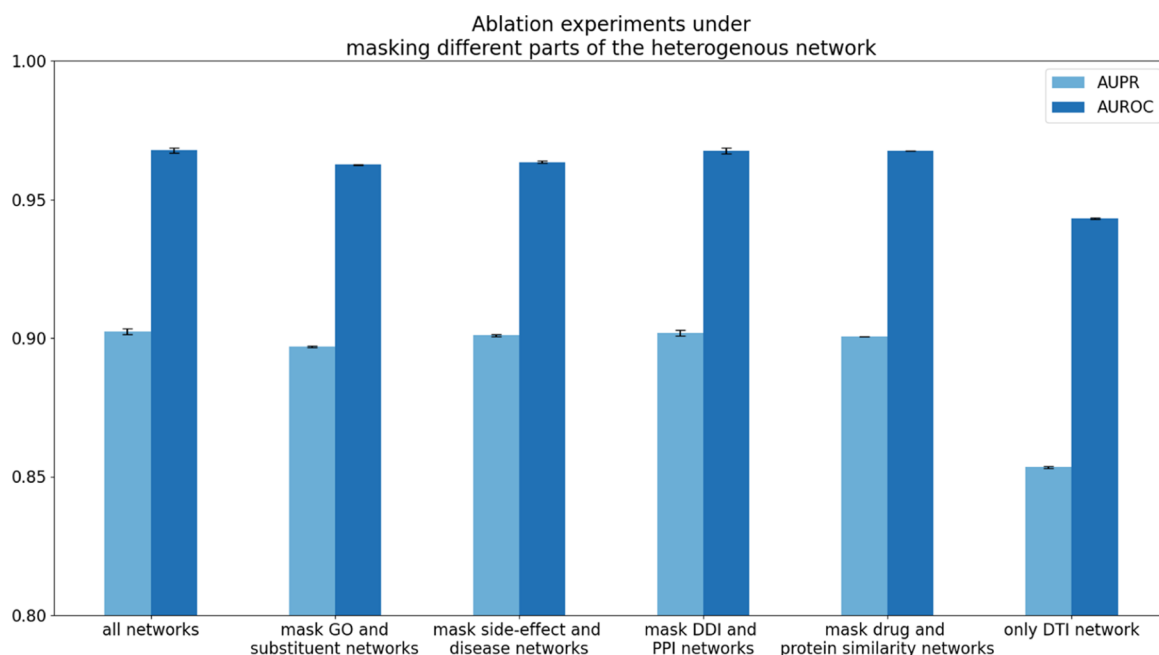


Figure 10. AUROC and AUPR of masking different subnetworks on D1923 with a 1:10 positive-to-negative sample ratio. Ablation studies were performed by masking various subnetworks related to drug or protein information to evaluate their contributions to the model's prediction performance.

dimension of 128, similar to the initialization strategy for disease and side effect nodes.

The results, as shown in Table 5, indicate that our initialization strategy achieved the best performance on the

Table 5. Results of MVCL-DTI with Different Initialization Strategies on D1923 with 1:10 Ratio

Different strategies	AUROC	AUPR
Random strategy	0.9479	0.8428
Luo's strategy	0.9522	0.8610
Yao's strategy	0.9558	0.8814
Our strategy	0.9677	0.9024

*Bold values in the table denote the highest performance.

DTI task. Compared to the next best baseline method, our model improved AUROC and AUPR by 1.19% and 2.1%, respectively. The main reason is that the newly added GO and substituents nodes provide additional functional information about proteins and drugs, making our strategy richer in information compared to the baselines.

3.9.2. Tuning Model Parameters of MVCL-DTI. Apart from protein and drug features, MVCL-DTI learns features from GO, substituents, side effect, and disease. It is easy to understand that those features, together with features from protein and drug, can help to determine the DTI in a qualitative way. However, how to fuse these features in a quantitative way is a hard task. The deep learning architecture of MVCL-DTI felicitously contributes to quantitatively solving this fusion task.

The quantitative fusion of all possible involved features for MVCL-DTI can be tuned in model hyperparameters at the training stage. The final model hyperparameters for MVCL-DTI can be found in Table S9. When determining the model structure, we performed a sensitivity analysis of the model's hyperparameters to find the optimal structure. First, we focused on determining the best number of layers for each view encoder.

As depicted in Figures 11 and S5, we found that increasing the number of layers beyond a certain point led to the vanishing gradient problem, which in turn caused a decline in model performance. This phenomenon occurs because deeper networks can make it challenging to propagate gradients back through the layers during training, leading to insufficient weight updates and suboptimal learning. Conversely, networks with a moderate number of layers were able to aggregate more informative features from the data, resulting in improved performance. After careful experimentation, we determined that a two-layer network struck the best balance, providing enough depth to capture complex interactions without suffering from gradient issues.

Additionally, we experimented with different dimensions for the hidden layers of our network. The hidden layer dimensions play a significant role in determining the capacity of the model to learn and represent complex patterns in the data. Larger dimensions allow the model to capture more detailed and nuanced information, but they also require more computational resources and can lead to overfitting if not managed properly. Due to resource constraints and the need to maintain a balance between model complexity and performance, we set the hidden layer dimensions to 2,048. This choice provided a sufficient capacity for the model to learn from the data effectively while keeping computational requirements within feasible limits.

4. CONCLUSIONS

In this study, we introduce MVCL-DTI, an innovative graph-based learning model designed to predict drug–target interactions (DTIs). MVCL-DTI begins by expanding the heterogeneous graph with the addition of two key nodes: Gene Ontology (GO) terms and substituents, along with new edges representing “Protein–GO Association” and “Drug–Substituents Association”. This expansion enriches the heterogeneous data, enabling our model to achieve a more comprehensive representation of biological networks. Subsequently, the model

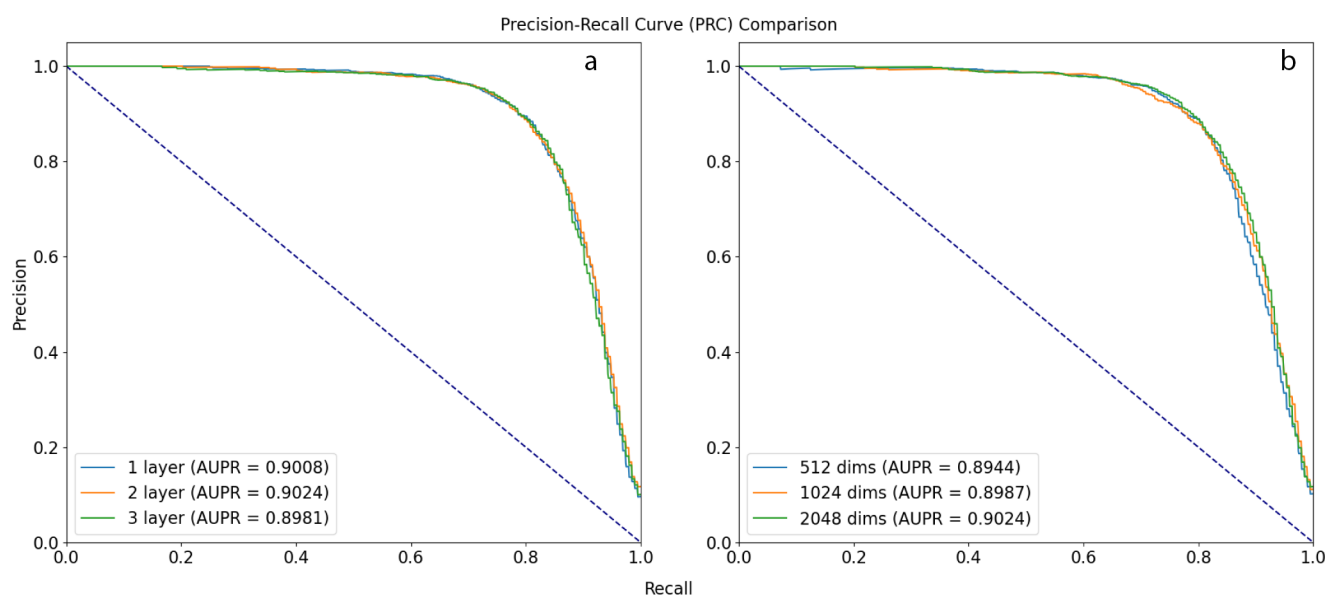


Figure 11. PR curve of MVCL-DTI with different layers and different dimensions. A sensitivity analysis of the model's hyperparameters is conducted to identify the optimal structure, focusing on (a) determining the ideal number of layers for each view encoder and (b) experimenting with different hidden layer dimensions within the network.

constructs three distinct views within the augmented heterogeneous graph: the neighbor view, the meta-path view, and the diffusion view. By leveraging contrastive learning, it effectively integrates network information from these different perspectives to capture multidimensional node representations with rich semantic content. Finally, the model reconstructs the original heterogeneous network to generate accurate DTI predictions.

To thoroughly analyze the performance of our model, we conducted an extensive array of experiments. Initially, leveraging a classic public data set, we employed diverse experimental configurations, including different positive-to-negative sample ratios, masking known DTIs with different ratios, masking redundant DTIs with various similarity metrics, and evaluating unique DTIs. The positive-to-negative sample ratio experiments highlighted the model's effectiveness in handling imbalanced data. Masking known DTIs confirmed its stability and generalization, even when redundant information was excluded. Evaluating unique DTIs showcased its capability in predicting new interactions without extensive prior knowledge. Despite the increased difficulty, MVCL-DTI excelled in both AUROC and AUPR metrics, demonstrating high precision and robustness. Then, the novel DTIs identified by MVCL-DTI, which are supported by literature, showcase the potential of MVCL-DTI to uncover new therapeutic targets and advance drugs discovery. Furthermore, case studies on COVID-19-related drugs highlighted MVCL-DTI's competitive performance in real-world applications, underscoring the practical utility of MVCL-DTI.

To uncover the reasons behind the superior performance of MVCL-DTI, we engaged in a comprehensive examination of its feature visualization and computational properties. By visually analyzing the evolution of protein features across the pre- and post-contrastive learning phases, it becomes evident that our model can capture semantically rich high-dimensional features. Our exploration of MVCL-DTI's computational properties yielded several key insights: 1) ESM2 and MACCS are the most effective representations for MVCL-DTI; 2) Integrating GO and substituent node feature representations substantially bolsters the model's efficacy; 3) Employing an adept

initialization strategy for the features trained by MVCL-DTI model is crucial for its enhanced predictive prowess. Ultimately, we elucidated to fine-tune the model's hyperparameters during the training phase to address the quantitative fusion of all possible involved features in MVCL-DTI. Thus, we can conclude that the robust performance of MVCL-DTI, as outlined in previous sections, is not a mere coincidence but a result of deliberate design and meticulous training of the model's computational components.

The success of MVCL-DTI can be attributed to several key factors: First, the incorporation of GO terms and substituents enriches the functional information on both proteins and drugs, significantly enhancing feature representations within the heterogeneous network. GO terms provide detailed annotations regarding proteins, such as their biological processes, molecular functions, and cellular components, which enrich the protein representations by introducing functional context. Similarly, substituents nodes contribute essential information about the chemical moieties of drugs, such as functional groups and structural modifications, which are key to determining their physicochemical properties and binding affinities. Second, the use of three distinct views captures multidimensional node representations, enabling a thorough understanding of the heterogeneous graph. Third, the contrastive learning module and the multiview attention-weighted fusion mechanism effectively integrate multiview representations. The contrastive learning module aligns and enhances the consistency of multiview representations, reducing redundancy and improving the robustness of feature representations. Meanwhile, the attention-weighted fusion module adaptively assigns weights to each view, enabling the model to prioritize the most informative perspectives dynamically. This synergy between contrastive learning and attention-weighted fusion ensures that the model can learn robust and meaningful features from heterogeneous data. Overall, while the MVCL-DTI model is primarily designed for DTI prediction, its methodology also holds the potential for broader application in similar scenarios in the future, such as drug–drug interactions and drug–disease

relationships, among other related fields. One of the key advantages of using a heterogeneous graph in DTI prediction is its inherent extensibility, which allows for seamless integration with other biological and biomedical data, facilitating multitask predictions across different domains. Additionally, the increasing evidence that miRNAs play critical roles in disease progression and their potential as drug targets offers an intriguing opportunity to extend our framework.⁵⁰ Computational models for predicting small molecule-miRNA associations (MMAs), such as JMSS-MMA⁵¹ and GAM-MDR,⁵² have demonstrated how graph-based methods can effectively explore the relationships between small molecules and miRNAs. Inspired by these approaches, we envision that our model could be adapted to integrate miRNA-related data within a heterogeneous graph framework, enabling the prediction of miRNA-small molecule interactions.

However, a limitation of our approach is its transductive nature: all information about the drugs and targets to be predicted must initially be present in the heterogeneous graph, meaning it cannot predict new drug-target interactions. In the future, we plan to address this by leveraging the heterogeneous graph as background knowledge, such as in a knowledge graph, to support predictions of novel drug-target interactions. By integrating dynamic and comprehensive heterogeneous graph information into knowledge graph frameworks, it becomes possible to infer interactions that are not explicitly present in the original graph, thus expanding the scope of the model to novel predictions.

■ ASSOCIATED CONTENT

Data Availability Statement

The source code is available at [<https://github.com/ljquanlab/MVCL-DTI>].

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jcim.4c02073>.

AUPR of different ratios of positive and negative samples (Figure S1); visualization of drug features, node-specific attention weights, and view-specific attention weights (Figures S2–S4); the ROC curve of MVCL-DTI with different layers and dimensions (Figure S5); AUC, AUPR, accuracy, recall, f1, and mcc of benchmarks with a 1:1/1:10 positive-to-negative sample ratio (Tables S1 and S2); the decrease in AUPR of MVCL-DTI (Table S3); pairwise statistical comparisons of AUROC and AUPR scores (Tables S4 and S5); AUROC and AUPR scores with mean, standard deviation, and 95% confidence intervals with a 1:1/1:10 positive-to-negative sample ratio (Tables S6 and S7); the targets we collected from DrugBank and pubchem (Table S8); hyperparameters settings for MVCL-DTI (Table S9) (PDF)

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Notes

The authors declare no competing financial interest.

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